A Dual Attention Neural Network for Airborne LiDAR Point Cloud Semantic Segmentation

Ka Zhang, Longjie Ye, Wen Xiao, Yehua Sheng, Shan Zhang, Xia Tao, and Yaqin Zhou

***Abstract*— With the development of airborne light detection and ranging (LiDAR) technology, it has become a common and efficient way to collect large-scale 3-D spatial information. However, efficient and automatic semantic segmentation of LiDAR data, in the form of 3-D point clouds, remains a persistent challenge. To address this, a dual attention neural network (DA-Net) is proposed, consisting of two different blocks, namely, augmented edge representation (AER) and elevation attentive pooling (EAP). First, the AER can adaptively represent local orientation and position, thereby effectively enhancing geometric information. Second, the captured local features of centroid points are utilized to further encode discriminative features using the EAP with the learned attention scores. Finally, a location homogeneity (LH) module is devised to explore the long-range relationship in an encoder–decoder network. Benefiting from the dual attention module, geometric information hidden in unorganized point clouds can be effectively propagated. Besides, the LH forces the network to pay attention to the semantic consistency of elevated objects, which facilitates both point- and object-level point cloud semantic segmentation for scene understanding. A benchmark dataset is used to assess the proposed method, which achieves an overall accuracy of 85.98% and an average F1 score of 72.31%. In addition, comparisons with other latest deep learning methods on the 2019 Data Fusion Contest dataset further demonstrate the robustness and generalization ability of the proposed method.**

***Index Terms*— Airborne light detection and ranging (LiDAR), dual attention neural network (DA-Net), location-homogeneity module, point cloud classification, Vaihingen dataset.**

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I. Introduction

COMPARED with oblique photogrammetry [1] and very high-resolution optical satellite images [2], airborne light detection and ranging (LiDAR) technology has the advantage of direct 3-D data acquisition with high precision and density. Nowadays, airborne LiDAR has been used in many surveying and geographic mapping applications, such as power corridor detection [3], [4], forest biomass estimation [5], 3-D modeling [6], and road extraction [7]. For these applications, semantic segmentation in terms of assigning a label to each 3-D point is a fundamental step in postprocessing. However, efficient semantic segmentation of airborne LiDAR point clouds remains a challenge due to the unordered and irregular nature of the data [8].

In previous years, many methods adopted classic machine learning classifiers and handcrafted features for point cloud semantic segmentation. These classifiers include support vector machine (SVM) [7], [9], random forest (RF) [10], [14], Markov random field (MRF) [11], conditional random field (CRF) [12], [13], expectation maximization (EM) [38], and AdaBoost [39]. Indeed, this type of method is likely to achieve satisfactory performance in certain scenes. However, feature engineering, referring to the creation of a set of handcrafted features, is often time-consuming. Besides, these methods often require prior knowledge of a given study area to choose an optimal classifier [14].

In recent years, deep learning methods, especially convolutional neural networks (CNNs), show the capacity to deal with challenging tasks, such as image classification [15], [16] and object detection [17], [18]. Although deep convolutional networks demonstrate outstanding performance in the field of 2-D computer vision, they cannot be directly applied to 3-D point clouds. Some researchers choose to convert point clouds to images or regular 3-D voxels before feeding them into CNN-like architecture [19], [20], [21]. The key to these methods is to transform such disordered, unstructured data into a regular grid array for easier processing, which, however, could potentially cause spatial information loss. Moreover, determining the projection angle or selecting the voxel resolution remains an issue. To completely solve the disorder and permutation invariance of point clouds, a point-based method called PointNet using the multilayer perceptron (MLP) and the max-pooling operation is proposed [22]. Though simple, PointNet served as a milestone that inspired many future studies. Researchers further devised various point-based network

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architectures, such as PointNet++ [23], PointCNN [24], and PointSift [25]. In the meantime, many benchmark datasets, such as S3DIS [26], ScanNet [27], Semantic 3D [28], and Urban 3D [29], became available for such supervised deep learning methods.

Recently, researchers endeavored to design novel point-based convolutions to further boost the performance in point cloud understanding. Some researchers utilize graph CNNs (GCNs) to capture the geometric information for point cloud semantic segmentation [30], [31], [32]. Meanwhile, some researchers tend to aggregate local features of point clouds by considering geometry structure [33], uneven density distribution [34], and global relationships [35]. Aiming to address the issue caused by overfitting, a method called Atrous XCRF integrating CRF similarity penalty into PointConv is proposed [36]. Although the above methods can learn rich geometric information, recognition of elevated objects is usually poor, not to mention the ineffective semantic representation of point cloud for 3-D scene understanding.

Feature extraction plays an essential role in the deep learning network architecture for point cloud understanding; however, there are still deficiencies in the related works of point cloud semantic segmentation. First, these networks lack the ability to represent robust geometric patterns of the local neighborhoods, which causes information loss in feature discrimination. Second, geometric characteristics of points merely learned from horizontal directions can be unreliable when extracting features during the encoding stage. Thus, it is desirable to integrate elevation information and distance discrepancy. Third, few studies focus on exploiting the long-range context, which has been proven useful for image semantic segmentation tasks. To address these issues, a dual attention neural network (DA-Net) is proposed for the semantic segmentation of airborne LiDAR point clouds. The main contributions of our research can be summarized as follows.

1) We develop an augmented edge representation (AER) block that can effectively learn rotation-invariant feature representation.
2) We propose an elevation attentive pooling (EAP) that can encode discriminate features based on geometric distance and the z coordinate using the attention mechanism.
3) We design a hierarchical network architecture that improves neighboring feature encoding capabilities and achieves effective semantic segmentation of large-scale point clouds.

II. Related Work

A. Traditional Machine Learning Methods

Early work on semantic segmentation of point clouds adopted a standard pipeline consisting of two steps: 1) learn a set of handcrafted features from a point's neighborhood and 2) select an appropriate classifier to predict the semantic label of each point. The key to feature extraction is to design suitable geometric descriptors, such as spin images [66], shape distributions [74], and signatures of histograms [75]. Considering that the above 3-D descriptors can be expensive to compute, some work [14], [76] used more general covariance matrix-based features, which can effectively represent a local geometric pattern. Furthermore, a variety of supervised classification methods (e.g., SVM, RF, and Adaboost) are applied for ALS point cloud semantic segmentation. However, due to the neglect of contextual information from neighborhoods, the segmentation results of these methods are often noisy, which manifests as label inconsistency [40]. To address this issue, a common strategy is to use individual point class possibility as a unary term in probabilistic graphical models, such as CRF and MRF [11], [13]. For example, Niemeyer *et al.* [41] designed a hierarchical point cloud classification model, which consists of a point-level CRF layer and a segment-level CRF layer. In their method, the point-level layer utilizes higher order cliques and geometrical features, while the segment-level layer operates on segment entities to incorporate a more extensive scale context. However, these methods have some limitations on computational efficiency and the recognition of multiscale objects in 3-D scenes [33].

With the recent advances in sparse coding theory, there is a push toward the representation of the higher level features [60], [61], [64]. A more compact feature representation can be obtained by using the encoding technique of the mid-level feature, such as the bag of visual words (BOW), the latent Dirichlet allocation (LDA), and the Fisher vector (FV). For instance, Kang and Yang [62] proposed a probabilistic graphical model that combines multiscale visual features and the LDA topic model for establishing robust semantic representation. Zhang *et al.* [63] first used a dictionary learning approach to represent features of multilevel point clusters. The discriminative features were further refined by the LDA topic model before employing multipath Adaboost classifiers. However, these methods suffer from a high computational burden when forming large dictionaries, which hinders the efficient semantic segmentation of point clouds.

B. Image-Based Deep Learning Methods

CNNs serve as a milestone, which enables deep learning to achieve outstanding results in 2-D computer vision. Motivated by this trend, many researchers turn to CNN for point cloud classification. However, point clouds cannot be directly fed into CNN. A natural idea is to project such data to collections of 2-D images so that point cloud classification can be regarded as pixelwise labeling. For example, Yang *et al.* [19] designed a multiscale CNN that integrates height above ground (HaG), intensity, flatness, sphericity, and angular deviation as the feature channel information for segment-based point cloud classification. Zhao *et al.* [20] constructed a set of multiscale contextual images containing various features (e.g., height, intensity, and roughness) and employed a CNN for training. Besides, they perform a tradeoff with results obtained from the bagged decision tree classifier to optimize the wrong classification of boundary points [20]. To achieve a more efficient data conversion, Rizaldy *et al.* [42] suggested converting point clouds into a panoramic image with multichannel (e.g., elevation, intensity, the number of echoes, and height

difference). Subsequently, an end-to-end full CNN was applied for pixelwise classification [42]. However, the transformation process from 3-D to 2-D inevitably brings about the loss of geometric information, which leads to misclassification, especially in complex scenes [8].

C. Voxel-Based Deep Learning Methods

The second primary strategy is to voxelize point clouds so that a 3-D CNN (e.g., 3-D convolution layer and 3-D pooling operation) can be used over regular grids. By converting 3-D point clouds to volumetric grids, the early work, VoxNet, achieved promising performance on point cloud classification [43]. Similarly, Huang and You [44] first transformed raw point clouds into voxels with a resolution of 0.3m. They then used LeNet-style 3-D convolution and max-pooling operations to construct a fully convolutional network structure for point cloud classification. Although excellent classification results have been achieved, such a voxel-based deep learning method is constrained by heavy memory consumption and computational cost. A powerful alternative is to apply an efficient data organization way, such as octree [21] and submanifold sparse convolution (SSC) [45]. Recently, Zhang *et al.* [63] proposed a voxel-based neural network, which can capture discriminative features in a hierarchical manner. However, these voxel-based methods suffer from spatial information loss and time-consuming problems [46]. Thus, it is desirable to devise a kind of neural network that can be directly used for the semantic segmentation of point clouds.

D. Point-Based Deep Learning Methods

Point-based methods deal with raw LiDAR data with a focus on learning the permutation invariance of points. Some of the earliest works aimed to design discrete or continuous point convolution operators for semantic segmentation of point clouds [8], [24], [47]. As a pioneer work of the above approaches, Qi *et al.* [22] designed a network architecture, namely, PointNet, which can directly consume the raw point cloud data in an end-to-end manner. While PointNet learned high-level pointwise features and global features through shared weight MLP and max-pooling operation, the derived results may be unsatisfactory since it cannot capture local structure information and contextual information. There are two techniques that can be used to improve the performance of point-based networks, neighboring feature aggregation, and attention mechanism.

Considering that current networks cannot learn sufficient information from the neighborhoods, some works propose a wide range of neighboring feature aggregation methods, such as hierarchical feature embedding [23] and orientation encoding [25]. Wen *et al.* [35] proposed a direction-constrained convolution operator that encodes the eight directions information of the 2-D receptive field to explore local representative features for point cloud semantic segmentation. Point clouds, as disordered data with a local topological structure, fit well with the nature of the graph. From the perspective of graph theory, researchers have formed two types of processing methods, including spatial domain methods [30], [48] and spectral domain methods [49], [50]. The spectral domain methods first convert the graph nodes to spectral signals in the Fourier frequency domain and then use the Laplacian eigenvalue to construct the graph convolution filterer. Zhang and Rabbat [51] designed a Graph-CNN model for point cloud classification, which is composed of two-layer spectral convolution hybrid multilevel resolution pooling and global pooling layers. Xu *et al.* [52] devised a family of graph convolution operators defined in the Fourier spectrum domain to capture local measurement line information. They used the multiplication of simple step functions and a Taylor polynomial to construct graph operation effectively. Since the learned spectral-domain convolution kernel is obtained from a fixed graph Laplacian matrix, the transferability is poor. With regards to spatial domain methods, Simonovsky and Komodakis [30] devised a graph convolution in which kernel weights are conditioned on edge labels for point clouds' classification. Wang *et al.* [53] built an edge convolution (Edge-Conv) operator based on the KNN graph to capture local geometric patterns from neighborhoods.

Recently, the self-attention mechanism has drawn significant interest due to its capability to encode long-range contextual features. Motivated by the transformer [77], various variants are proposed for enhancing the deep discriminative features of point clouds that are proposed, such as point attention transformers (PATs) [67] and Point2Sequence [68]. Some work on the point convolution operators has also been tried. For example, Feng *et al.* [56] proposed a local attention-Edge-Conv, which aggregates the point features as a weighted sum of its neighbors. Zhang *et al.* [69] proposed a relation-aware global attention module to learn the global structural information. Xie *et al.* [70] proposed an attention-based graph convolutional network to explore the relationship between local neighborhoods. Despite remarkable performance on indoor datasets, few works attempt to employ the attention mechanism for semantic segmentation of outdoor point clouds. Considering that the existing convolution operators cannot capture geometric features from the neighborhood, Li *et al.* [33] proposed a dense hierarchical network that consists of geometric attention convolution and elevation attention units. Li *et al.* [34] designed a density-aware convolution that uses the pointwise density to adjust the learned convolution kernel parameters. In particular, Wen *et al.* [54] propose a graph attention CNN composed of the global attention module and the local graph attention module to realize semantic segmentation of airborne LiDAR point clouds. Accordingly, these methods either use global attention on the local features or share dot-product attention weights with different feature channels [71], [72], [73].

Compared with traditional machine learning methods, these deep learning methods have advantages in automatic workflow and effective extraction of discriminative features for semantic segmentation of airborne LiDAR point clouds. However, since most deep learning-based methods are designed for indoor point clouds, they have limitations in discriminative feature

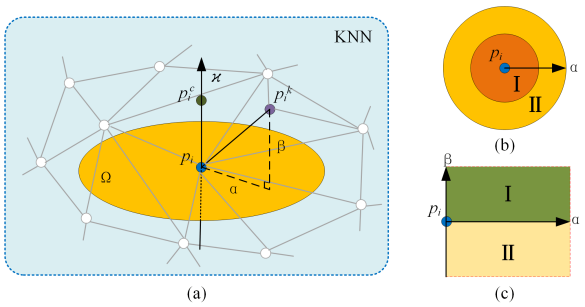


Fig. 1. Illustration of modeling orientation features. (a) Cylindrical reference frame including origin point p_i and centroidal vector $\mathbb{M}$, and the relative coordinate α and β. (b) Top view of coordinate system division. (c) Side view of coordinate system division.

learning and poor results of semantic segmentation in large-scale scenes.

III. Methodology

In this section, details of DA-Net for point cloud classification are described. First, the DA module is introduced, which consists of two different blocks, AER, and EAP. Then, a location homogeneity (LH) module is explained to encode long-range relationships. Finally, an encoder–decoder network architecture based on the DA and the LH is presented.

A. Dual Attention Module

Existing studies often apply max pooling to aggregate information from neighboring, leading to the majority of geometric patterns being dropped. In contrast, the DA containing an attention mechanism is proposed to effectively learn essential features without information loss. In this section, we give the details of the DA and illustrate its architecture.

1) Augmented Edge Representation: The adjacency information from different objects is of great significance, which means that local features directly learned from 3-D point clouds can benefit from such information. However, few studies have been conducted to further investigate the effective representation of adjacency information, also referred to as edges. In this case, we propose an AER to encode local spatial features that combine multidimensional orientation and relative position (see Fig. 1). Note that the spatial representation from the local region is represented in a spherical coordinate system.

Given a point p_i, its neighboring points $\{p_i^1, p_i^2, \ldots, p_i^k\}$ are first gathered by the K-nearest neighbors (KNNs) algorithm based on the pointwise Euclidean distance. When using the KNN algorithm to find appropriate neighbors, a fixed value K needs to be selected (e.g., 8, 16, 32, and 48). In this way, we first calculate edge feature vector e_i^k and distance vector dis_i^k, and then, relative position ε_i^k can be expressed as follows:

$$\varepsilon_i^k = \left[e_i^k, \text{dis}_i^k\right] = \left[\left(p_i - p_i^k\right), \left\|p_i - p_i^k\right\|\right] \tag{1}$$

where [] denotes the concatenation operation; || || is the Euclidean distance between p_i and its neighboring point p_i^k. For multidimensional orientation, we define a vector $o_i^k \in \mathbb{R}^4$ to represent the direction from p_i to p_i^k. To model local geometrical information of each edge, a cylindrical reference frame is introduced, as shown in Fig. 1(a). Inspired by [66], an oriented coordinate system (p_i, $\mathbb{M}$) that uses plane Ω through p_i oriented perpendicularly to centroidal vector $\mathbb{M}$ is defined. To obtain $\mathbb{M}$, the centroid p_i^c of the local neighborhood (i.e., KNN) is introduced. Then, $\mathbb{M}$ is defined based on the unitized direction vector from p_i to p_i^c. Given a point p_i^k, the coordinate value α is defined as the perpendicular distance to $\mathbb{M}$, and β is defined as the signed perpendicular distance to Ω as follows:

$$\begin{cases} \alpha = \sqrt{\left\|p_i^k - p_i\right\|^2 - \left(\mathbb{M} \cdot \left(p_i^k - p_i\right)\right)^2} \\ \beta = \mathbb{M} \cdot \left(p_i^k - p_i\right). \end{cases} \tag{2}$$

In order to form o_i^k, we utilize the partition strategy to partition a feature space into four bins based on the cylindrical reference frame. As shown in Fig. 1(b) and (c), the 2-D space is first partitioned into $S_a = 2$ areas using half of $\alpha_{\max}$ along the plane Ω. Each sector is further divided into $S_z = 2$ zones according to the sign of β along the centroidal vector $\mathbb{M}$. It is worth noting that, using the partition strategy, o_i^k is represented as a binary vector with a length of $S_a \times S_z$. In other words, a value of 1 corresponds to the located region in the cylindrical reference frame. Overall, this kind of local representation has two distinctive benefits: 1) the cylindrical reference frame enables us to learn rotation-invariant feature representation and 2) the centroidal vector $\mathbb{M}$ can reduce the effect of some disturbances, such as occlusion and sampling due to the robustness of the centroid of the local neighborhood.

For each neighboring point p_i^k with its corresponding feature f_i^k, an aggregated edge feature, $\dot{f}_i^k$, can be computed as follows:

$$\dot{f}_i^k = \varkappa\left(\left[\varepsilon_i^k, o_i^k, f_i^k\right]\right) \tag{3}$$

where $\varkappa$ denotes an MLP with a batch normalized layer and an ReLu activation function; [] denotes the concatenation operation.

2) Elevation Attentive Pooling: Given the gathered KNN feature $\dot{F}_i\{\dot{f}_i^1, \dot{f}_i^2, \ldots, \dot{f}_i^k\}$, it is important to design an ensemble block for integrating such features. Meanwhile, extensive work has proved that elevation information derived from LiDAR data plays an important role in the classification task. Instead of using the attention weights directly generated from the z coordinate to adjust the final feature map per channel [33], we aim to extract effective feature representation in the local neighborhood based on geometric distance and the z coordinate. Therefore, an EAP block is proposed to adaptively aggregate the $\dot{F}_i$ using the attention mechanism.

As illustrated in Fig. 2, to form aggregated features, the EAP first performs dot product on z coordinate in neighboring points and relative distance computed by 3-D coordinates based on the Euclidean distance. Specifically, the encoded elevation feature, $\mathfrak{h}_i^k$, can be expressed as follows:

$$\mathfrak{h}_i^k = z_i^k \cdot \exp\left(-\text{dis}_i^k\right) \tag{4}$$

$$\text{dis}_i^k = \sqrt{\left(x_i^k\right)^2 + \left(y_i^k\right)^2 + \left(z_i^k\right)^2} \tag{5}$$

where "·" denotes the dot product; x_i^k, y_i^k, and z_i^k are the relative coordinate in the Cartesian coordinate system. Meanwhile,

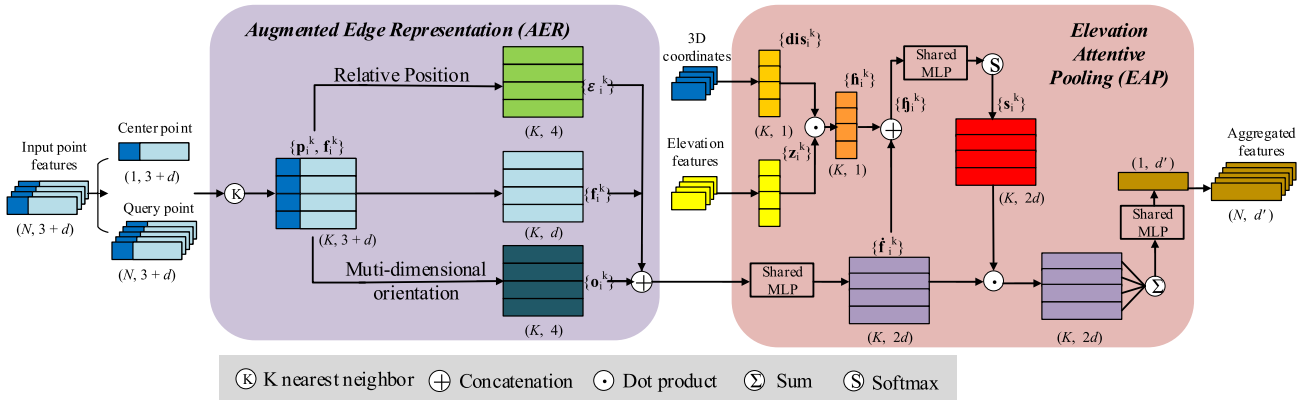


Fig. 2. Architecture of the proposed DA module.

the output of AER, $\dot{f}_i^k$, is concatenated with $\mathfrak{h}_i^k$ in order to generate a new feature vector $\mathfrak{h}_i^k$ as follows:

$$\mathfrak{h}_i^k = \left[\dot{f}_i^k, \mathfrak{h}_i^k\right] \tag{6}$$

where [] denotes the concatenation operation. In addition, a shared MLP denoted as $\varkappa$ is used to form the attentive scores. Followed by a softmax function, the attentive scores s_i^k can be computed as follows:

$$s_i^k = \text{softmax}\left(\varkappa\left(\mathfrak{h}_i^k\right)\right). \tag{7}$$

Eventually, the aggregated features, f_i^{new} can be calculated via the learned attentive scores s_i^K using the weighted sum operation

$$f_i^{\text{new}} = \sum_i^K s_i^k \cdot \dot{f}_i^k. \tag{8}$$

B. Location Homogeneity Module

In a traditional CNN network, the output feature map of each convolution layer has strong representation power for the local pattern. However, the convolution kernel still cannot provide long-range information between different local receptive fields due to the fixed local receptive fields. In other words, the long-range relationship between different neighborhoods cannot be propagated. This will result in poor detection of elevated objects, particularly trees and roofs, due to the ignoring of spatial dependencies. This phenomenon will affect the consistency of point cloud segmentation in the entire scene, which is manifested as a decrease in the accuracy of the final results.

Inspired by [55] and [56], an LH module is developed to exploit the long-range dependencies. Given the input pointwise feature $\boldsymbol{F}_{\text{in}} \in \mathbb{R}^{N\times d}$, three new features $\boldsymbol{F}_A$, $\boldsymbol{F}_B$, and $\boldsymbol{F}_C \in \mathbb{R}^{N\times d}$ are calculated via an MLP layer, respectively. Then, matrix multiplication is applied between $\boldsymbol{F}_B$ and the transpose of $\boldsymbol{F}_A$. The location attention map $\boldsymbol{A} \in \mathbb{R}^{N\times N}$ is finally obtained by using a softmax function to the output of matrix multiplication. It can be defined as (6)

$$a_{ij} = \frac{\exp\left(\boldsymbol{F}_{B_i} \cdot \boldsymbol{F}_{A_j}\right)}{\sqrt{d} \times \sum_{j=1}^{N} \exp\left(\boldsymbol{F}_{B_i} \cdot \boldsymbol{F}_{A_j}\right)} \tag{9}$$

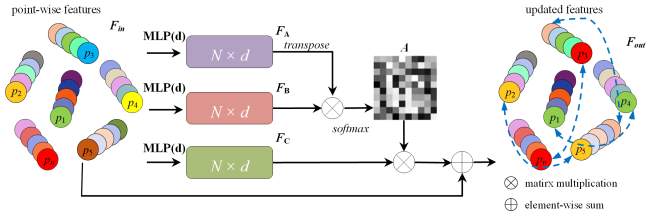


Fig. 3. Proposed LH module. A collection of colored circles represents the input pointwise features $\boldsymbol{F}_{\text{in}}$. The purple, pink, and green rectangles represent FA, FB, and FC mapped by an MLP layer, respectively. The gray mosaic image is the position attention map A. The blue dotted arrow indicates long-range dependencies. Finally, the output of the LH is $\boldsymbol{F}_{\text{out}}$. N denotes the number of point clouds, and d represents the dimension of the feature.

where i and j represent the point order of $\boldsymbol{F}_B$ and $\boldsymbol{F}_A$, respectively, and a_{ij} denotes the ith locations' influence on the jth location of $\boldsymbol{A}$. To prevent the softmax function from making the gradient of A smooth, it is divided by the root of dimension number d. Generally, the closer the feature vectors of two positions, the stronger the interaction between them.

Thereafter, another matrix multiplication of $\boldsymbol{A}$ and $\boldsymbol{F}_C$ is performed. In addition, a scale parameter μ is applied to progressively increase the weight of long-range dependencies. Thus, for a point cloud $\mathbf{P}$, the augmented feature $\boldsymbol{F}_{\text{out}} \in \mathbb{R}^{N\times d}$ can be formulated as follows:

$$\boldsymbol{F}_{\text{out}} = \mu(\boldsymbol{A} \cdot \boldsymbol{F}_C) + \boldsymbol{F}_{\text{in}} \tag{10}$$

where the operation "+" denotes the elementwise sum and the operation "·" is the matrix multiplication. To summarize, the LH encodes the long-range dependencies to aggregate geometric features. Specifically, after delivering such information (blue dotted arrow in Fig. 3), a point can automatically capture the important geometric patterns coming from p_n. As depicted in Fig. 3, p_3 receives features from p_n, thereby enriching its semantic representation.

C. Network Architecture

To enable our network to learn hierarchical features, two advanced technologies introduced by PointNet++, including the farthest point sampling (FPS) and inverse distance weight (IDW) interpolation, are adopted to construct the encoder–decoder architecture. The network architecture is

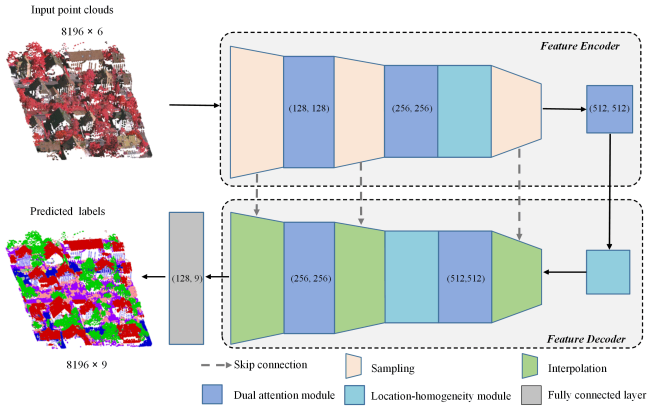


Fig. 4. Schematic of the proposed DA-Net architecture. The network consists of two parts (e.g., a feature encoder and a feature decoder). In the encoding stage, along with sampling, the DA module and the location-homogeneity module are recursively used to generate hierarchical features. In the decoding stage, we interpolate the encoded features and employ a skip connection for feature propagation. Finally, the label of each point is predicted to pass through the fully connected layer. The input of the network is 8192 × 6, which means 8192 points with six feature dimensions (x, y, z, R, G, B) for each point; the output of the network is the predicted label of each point.

similar to the U-shape, an end-to-end network architecture with sampling and interpolation, as shown in Fig. 4. Due to the unbalanced number of categories, a rebalancing strategy is adopted for the weight of the loss function. Specifically, the weight coefficient of each category is calculated by the inverse ratio of the number of points of each category to the total point number. The calculation formula is given as follows:

$$W_c = \frac{1}{\ln\left(\alpha + \frac{N_c}{\sum_{c=1}^{C} N_c}\right)} \tag{11}$$

where α is the balance coefficient, 1.2 in default; N_C represents the total number of points of the cth category; and C represents the total number of categories. Furthermore, we use the weighted cross-entropy loss function in the training process of the model. The formula can be expressed as follows:

$$\text{Loss} = -\sum_{c=1}^{C} W_c \times (y_c \log \hat{y}_c + (1 - y_c)\log(1 - \hat{y}_c)) \tag{12}$$

where y_c and $\hat{y}_c$, respectively, represent the probability that the ground truth label and predicted label of the cth sample are class c; W_c represents the rebalancing weight of the cth class.

In the encoding stage, the sampling layer reduces the number of points to 1024, 256, and 64. The corresponding output channel feature numbers are 128, 256, and 512, respectively. In the decoding stage, the feature propagation layer restores the number of points to 256, 1024, and 8192. The proposed dual attention (DA) module and the location-homogeneity module are recursively embedded behind the downsampling layer and the upsampling layer. Finally, a fully connected layer is used to complete the pointwise label prediction. It is worth noting that the fully connected layer is followed by a softmax function to map the feature to a reasonable probability distribution, and the class with the highest probability is regarded as the predicted label. In addition, the skip connection is applied in the corresponding sampling layer to maintain low-level feature information. This technology has been proven to be effective in improving the performance of the model in the classic CNN [15]. In particular, low-level features have a smaller receptive field, while features located at a higher level have a larger receptive field. By stacking features from different levels, better results can be achieved in point cloud semantic segmentation.

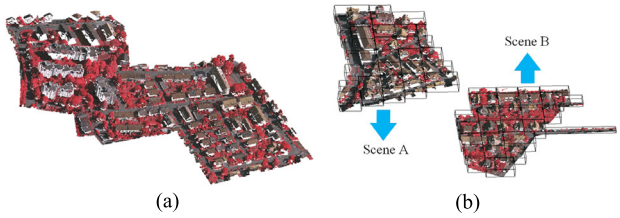


Fig. 5. Vaihingen 3-D labeling dataset displayed in false color (IR: R; R: G; and G: B). (a) Training set. (b) Test set. Test data include two scenes, namely, Scene A and Scene B.

IV. Experiment

In this section, two airborne LiDAR datasets, namely, the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen benchmark dataset and the 2019 Data Fusion Contest (DFC) dataset, are used to test the performance of the proposed DA-Net. In addition, we evaluate the effectiveness of the DA module and the LH module through a series of ablation studies on the ISPRS Vaihingen dataset. The proposed model and other commonly used deep learning methods are compared on both airborne LiDAR datasets.

A. Experiments on ISPRS Vaihingen Dataset

The ISPRS Vaihingen dataset[1] obtained by the Leica ALS50 system in Germany is used to evaluate the proposed method [13]. The information of each point is given by {x, y, z, intensity, number of echoes, total number of echoes}. In addition, a multispectral orthophoto image with a resolution of 0.02 m corresponding to IR-R-G in this area is provided. By calculating the band value of the corresponding pixel in the raster image corresponding to each point, the corresponding color information can be obtained, as shown in Fig. 5(a). In particular, we use spectral features while training our network. The dataset has already been divided into a training set and a test set by the organizer. The training set contains 753 876 points, and the test set contains 411 722 points. They are classified into nine categories, including powerline, low vegetation, impervious surfaces, car, fence/hedge, roof, facade, shrub, and tree. Table I explicitly introduces the number of points in each category in the training set and the test set.

1) Preprocessing and Evaluation Criteria: For the preprocessing of training data, a method similar to [35] is used. During the training process, we randomly select a 30 m × 30 m block in the training sample. Similar to

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/3d-semantic-labeling.aspx

TABLE I
SAMPLE DISTRIBUTION FOR EACH CATEGORY OF THE VAIHINGEN DATASET

Class	Training data	Test data
Powerline	546	600
Low vegetation	180850	98690
Impervious surfaces	193723	101986
Car	4614	3708
Fence/Hedge	12070	7422
Roof	152045	109048
Facade	27250	11224
Shrub	47605	24818
Tree	135173	54226
Sum	753876	411722

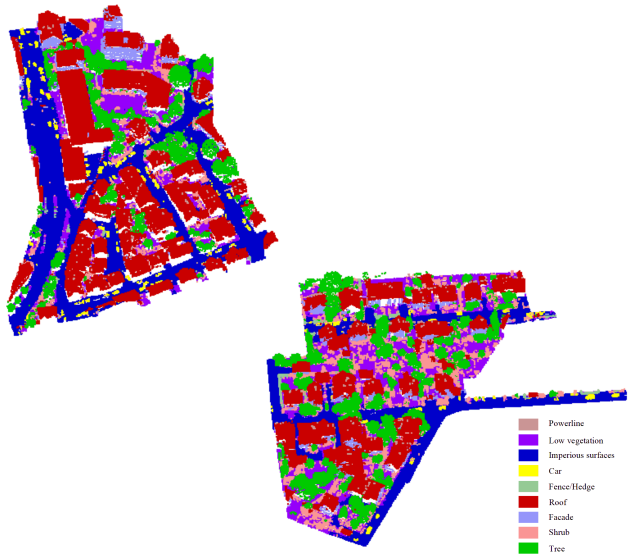


Fig. 6. Classification results of our proposed DA-Net on the Vaihingen dataset. Colors represent different categories.

PointNet, we randomly sample 8192 points in the block as input to the model. If the number of point clouds in each block is less than 8192, we perform replacement and sampling. To prevent overfitting, we adopt a technique with a random dropping rate of 12.5% for the selected 8192 points during training. At the same time, to boost the performance during training, we translate the coordinates of each block to the local coordinate system by subtracting the coordinates of the left bottom corner. Finally, we perform random rotations along the z-axis to enhance the orientation invariance. In the test stage, the dataset is first divided into two zones, namely, Scene A and Scene B due to the discrete distribution of the data. Then, both Scene A and Scene B are further sliced into 30 m $\times$ 30 m blocks without overlap. After the partition procedure, 76 blocks are obtained. Note that some of these blocks cover a long and narrow area due to the irregular shape of the test data. To ensure the integrity of point clouds in each block, we merge them into 58 complete blocks [see Fig 5(b)]. Finally, 8192 points are sampled in each block and translated to the local coordinate system as in the training stage.

Four common metrics for evaluation are used, including overall accuracy (OA), precision, recall, and F1 score. Specifically, OA means the percentage of correctly classified points out of the total number of points. The formula is given as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{13}$$

where TP, TN, FP, and FN, respectively, represent true positive, true negative, false positive, and false negative. In detail, TP represents the number of positive samples correctly classified by the classifier, TN represents the number of negative samples correctly classified by the classifier, FP represents the number of false-positive samples, and FN represents the number of false-negative samples.

The F1 score is computed from the precision and recall of each class. F1 scores can be better than OA in certain situations where the number of categories is not balanced. The calculation formula is given as follows:

$$\mathrm{F1} = 2 \times \frac{\mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} \tag{14}$$

in which the precision rate and the recall rate are calculated as follows:

$$\begin{cases} \mathrm{precision} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP}) \\ \mathrm{recall} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}). \end{cases} \tag{15}$$

2) Implementation Details: The proposed model is implemented using the framework of Tensorflow 1.14.0. We use an RTX 2080 Ti GPU to train the network model, which took about 10 h. During the training stage, the Adam optimizer is used to optimize the model with an initial learning rate of 0.005. We employ the cosine decay measurement to adjust the learning rate, and the number of decay steps is 30,000. The batch size, the decay rate, and the max epoch are set to 6, 0.5, and 1000. In our encoder–decoder network, we set the number of neighbor points for upsampling and downsampling to 32. In the construction of the KNN graph, we use the setting of $K = 16$. The model parameters configured are derived from a series of comparative experiments.

3) Classification Results: To qualitatively evaluate the classification results of the proposed model, we show the classification results and the classification error map in Figs. 6 and 7. As shown in Fig. 6, it seems that the categories of most points in the test data scene can be correctly identified. As depicted in Fig. 7, the major misclassified points appear at the border between hedges and shrubs. In particular, our model can accurately distinguish between roof and tree categories even though they are similar in height and have no apparent boundaries.

In addition, the quantitative results are displayed in a confusion matrix (see Table II), which shows that the F1 scores of five out of the nine categories are more than 75%. The categories with better classification accuracy are low

TABLE II
CONFUSION MATRIX OF THE PROPOSED DA-NET ON THE VAIHINGEN DATASET WITH PER-CLASS PRECISION, RECALL RATE, AND F1 SCORE (%)

Categories	Powerline	Low vegetation	Impervious surfaces	Car	Fence/Hedge	Roof	Facade	Shrub	Tree
Powerline	403	2	0	0	0	117	2	1	75
Low vegetation	0	81491	8671	346	173	968	294	4144	2603
Impervious surfaces	0	4751	96801	72	2	150	42	144	24
Car	0	263	117	2908	26	123	45	203	23
Fence/Hedge	0	885	155	62	1858	316	169	2713	1264
Roof	134	714	72	8	2	105079	1851	438	750
Facade	12	926	153	68	15	2010	6629	878	533
Shrub	0	4479	149	142	302	1328	614	11244	6560
Tree	8	1195	10	64	138	996	590	3640	47585
Prcision	72.35	86.05	91.21	79.24	73.85	94.59	64.76	48.04	80.09
Recall	67.17	82.57	94.92	78.43	25.03	96.36	59.06	45.31	87.75
F1 score	69.66	84.27	93.03	78.83	37.39	95.47	61.78	46.63	83.74

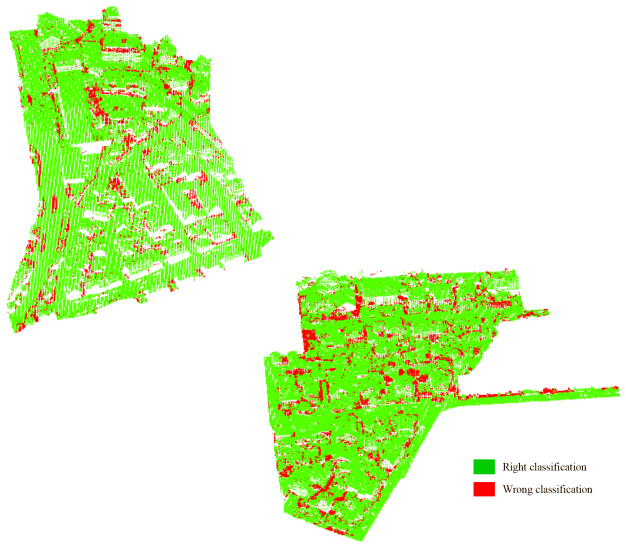


Fig. 7. Error map of the proposed DA-Net on the Vaihingen dataset. The green points and red points represent the correct and wrong segmentation results, respectively.

TABLE III
QUANTITATIVE RESULTS (%) FOR DA MODULE WITH DIFFERENT SIZES OF NEIGHBORHOOD ON THE VAIHINGEN DATASET

CLASS	K=8	K=16	K=32	K=48
POWERLINE	60.7	69.7	67.9	66.6
LOW VEGETATION	81.4	84.3	84.0	83.5
IMPERVIOUS SURFACES	90.8	93.0	92.6	92.2
CAR	72.9	78.8	77.1	74.7
FENCE/HEDGE	38.8	37.4	35.7	31.0
ROOF	94.7	95.5	95.3	95.2
FACADE	58.0	61.8	61.1	59.8
SHRUB	46.2	46.6	45.7	43.6
TREE	83.1	83.7	83.0	82.3
AVG.F1	69.6	72.3	71.4	69.9

vegetation, impervious surfaces, cars, roofs, and trees. It is worth noting that the proposed method achieves impressive results on elevated objects, especially the trees, which have a severe overlaying issue (e.g., trees grow close to the facade). Although the number of points of the powerline and the car is relatively small, our method still achieves satisfactory results, as shown in Fig. 8.

On the other hand, the proposed method performs less effectively for the fence and shrub categories due to their similar height distribution and shape pattern. According to the confusion matrix, a considerable amount of points belonging to trees are classified as shrubs. It is because the two types have similar geometric patterns and geographical distribution.

Furthermore, Fig. 9 shows the pointwise features obtained from the fully connected layer of DA-Net. By observing one-channel feature maps of a sampled point cloud, it can be seen that some channels have a greater response to the height and gradient features [see Fig. 9(a)]. On the other hand, thanks to the introduced spectral features, roofs, impervious surfaces, and facades can be effectively identified. The line graphs in Fig. 9(b) present the 128-D features of points of each category. Accordingly, it can be concluded that the proposed DA-Net can capture high-level discriminative features for the semantic segmentation of point clouds. These discriminative features enable our method to work robustly in complex scenarios.

4) Ablation Study:

a) Sensitives of parameters: In order to select suitable neighbors for the DA module, we conduct experiments to test the effects of different sizes of the neighborhood on classification results. The K value is set as 8, 16, 32, and 48. In Table III, it can be seen that, when K is set as 16, our method achieves the best result. When K is greater than 16, the avg F1 score decreases significantly as the parameters of the DA module increase. In this case, redundant information

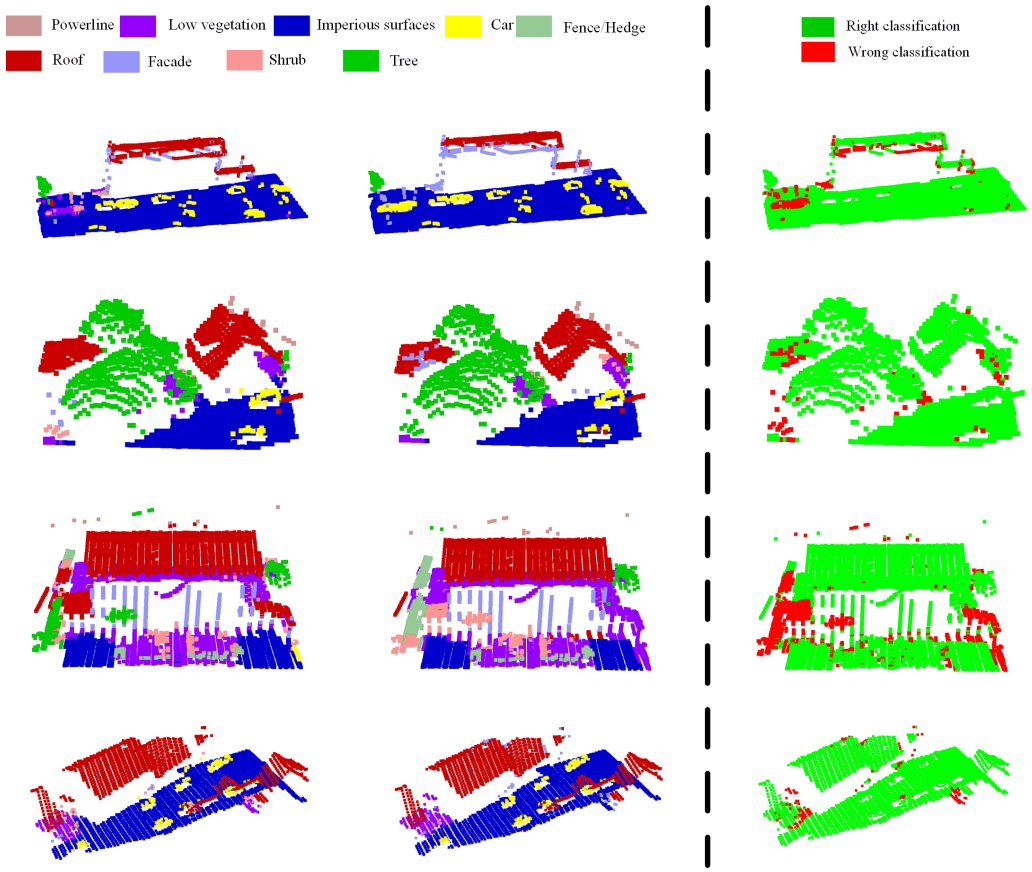


Fig. 8. Details of results on the Vaihingen dataset.

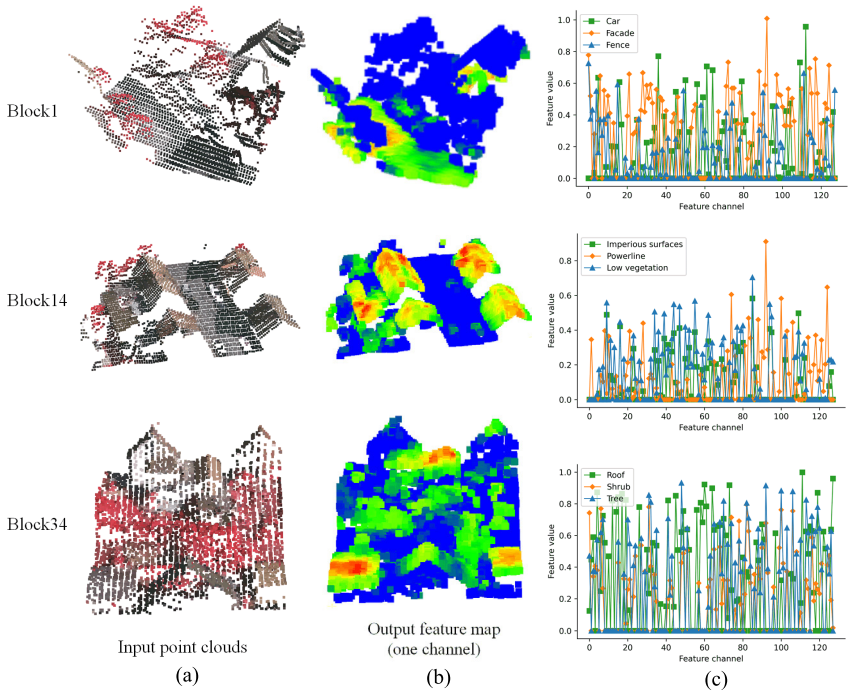


Fig. 9. Visualizing the feature line graphs and feature maps learned from the fully connected layer of DA-Net. (a) Point clouds of a selected block displayed in false color (IR: R; R: G; and G: B). (b) One-channel feature map of the corresponding point clouds; red represents the higher response, and blue represents the lower response. (c) Feature line charts.

from local neighbors will be learned, which has a negative effect on classification results.

b) Effectiveness of the DA module: To further investigate the effectiveness of the proposed DA module, three variant

TABLE IV
QUANTITATIVE RESULTS (%) FOR DA MODULE WITH DIFFERENT COMPONENTS ON THE VAIHINGEN DATASET

Class	Baseline	DA-Net/ AER	DA-Net/ EAP	DA-Net
Powerline	61.2	68.9	69.9	69.7
Low vegetation	82.9	84.5	83.7	84.3
Impervious surfaces	91.0	93.0	92.8	93.0
Car	74.1	77.0	74.3	78.8
Fence/Hedge	33.0	35.9	42.2	37.4
Roof	95.0	95.4	95.3	95.5
Facade	58.4	60.6	59.9	61.8
Shrub	44.2	46.1	44.0	46.6
Tree	82.4	83.0	83.6	83.7
Avg.F1	69.1	71.6	71.7	72.3

TABLE V
QUANTITATIVE RESULTS (%) FOR DA-NET WITH AND WITHOUT LH MODULE ON THE VAIHINGEN DATASET

Class	DA-Net without LH	DA-Net
Powerline	68.2	**69.7**
Low vegetation	84.2	**84.3**
Impervious surfaces	92.9	**93.0**
Car	76.2	**78.8**
Fence/Hedge	**40.5**	37.4
Roof	95.4	**95.5**
Facade	60.7	**61.8**
Shrub	45.6	46.6
Tree	83.6	**83.7**
Avg.F1	71.9	**72.3**

models are designed to conduct ablation studies: 1) DA-Net without the DA module, which is replaced by MLP (baseline); 2) DA-Net with only AER, i.e., the EAP in the DA module is replaced by MLP (DA-Net/AER); and 3) DA-Net with only EAP, i.e., the AER is replaced by MLP (DA-Net/EAP). The sizes of the neighborhood in these models are set as 16 by default. Table IV shows the quantitative results of the above three models. We can see that adding the AER enables the baseline to gain 2.5% in the average F1 score, while applying the EAP to the network can increase the average F1 score from 69.1% to 71.7%. Furthermore, by combining two blocks, the proposed DA-Net achieves the best performance with an average F1 score of 72.3%, showing that the introduction of the DA module can improve the robustness of local representation.

c) Effectiveness of the LH module: We further test the effectiveness of the LH module on classification results, as shown in Table V. DA-Net without LH means that the LH module in the network is removed without any replacement at the corresponding place. The results show that adding the LH module helps to boost the classification performance. Specifically, most categories benefit from the introduction of the LH module in terms of the F1 score. This is because the LH helps the DA-Net take the long-range dependency into account, thereby correcting some potential classification errors.

5) Comparisons With Other Methods: To further assess the proposed DA-Net, we compare it with the other eight models in the ISPRS 3-D labeling contest. These models include UM [57], WHUY2, WHUY3 [58], LUH [41], BIJ_W [59], RIT_1 [8], NANJ2 [20], and WhuY4 [19]. F1 scores of each category of these models are displayed in Table VI, in which OA and average F1 scores are shown in the last two columns. As indicated in Table VI, the proposed method outperforms all the other methods in terms of OA and average F1 score. Specifically, the OA and average F1 scores are 0.7% and 3.0% higher than the best of other methods, respectively. Note that it achieves state-of-the-art results in six categories (e.g., powerline, impervious surfaces, car, roof, facade, and tree).

In comparison, the NANJ2 [20] achieves the best results in the two categories of low vegetation and shrub. WHUY4 [19] is 17.4% higher than ours in terms of the F1 score of fence/hedge. However, it did not achieve the best performance in terms of OA and average F1 score. It seems that these methods tend to precisely classify a specific category rather than all categories, resulting in inferior performance on the average F1. One can find that our method can produce more consistent results across most categories, as the weighted loss is applied to deal with the imbalance of the data. On the other hand, the proposed DA-Net has achieved excellent results in categories with significant elevation differences, such as powerline, roof, and car. One possible reason is that the EAP amplifies the difference in height distribution and, therefore, improves the recognition ability.

Furthermore, the proposed DA-Net is compared with other advanced deep learning models, as shown in Table VII. They include PointNet++ [23], PointSIFT [25], PointCNN [24], the A-XCRF model [36], KPConv [47], RandLA-Net [78], DGCNN [53], GADH-Net [33], GAC-Net [32], and GACNN [54]. Note that OA and average F1 scores are also listed in the last two columns. As illustrated in the table, the proposed DA-Net has the highest F1 score in five categories (low vegetation, impervious surfaces, car, roof, and tree), outperforming other models by 1.6%, 1.1%, 1.0%, 1.0%, and 1.0%, respectively. The F1 score of our method in the powerline is slightly lower than GACNN [54]. A possible reason is that the method introduces a density attention unit, which benefits this category greatly. The A-XCRF model [36] adopts a postprocessing technology to deal with overfitting; thus, it can achieve a better F1 score in the two categories of fence/hedge and shrub, which are difficult to distinguish. The GADH-Net [33] improves the classification result of fence/hedge (44.2%) by learning discriminative geometry representation. Nevertheless, the proposed DA-Net still achieves the best results in terms of the OA (85.9%), which validates the effectiveness of the DA and the LH.

As shown in Fig. 10, qualitative comparisons of the LUH [41], NANJ2 [20], WhuY4 [19], DANCE-NET [34],

TABLE VI

QUANTITATIVE RESULTS OF THE PROPOSED DA-NET AND OTHER EIGHT METHODS ON THE VAIHINGEN DATASET (%)

Categories	Powerline	Low vegetation	Impervious surfaces	Car	Fence/Hedge	Roof	Facade	Shrub	Tree	OA	Avg.F1
UM	46.1	79.0	89.1	47.7	5.2	92.0	52.7	40.9	77.9	80.8	59.0
WhuY2	31.9	80.0	88.9	40.8	24.5	93.1	49.4	41.1	77.3	81.0	58.6
WhuY3	37.1	81.4	90.1	63.4	23.9	93.4	47.5	39.9	78.0	82.3	61.6
LUH	59.6	77.5	91.1	73.1	34.0	94.2	56.3	46.6	83.1	81.6	68.4
BIJ_W	13.8	78.5	90.5	56.4	36.3	92.2	53.2	43.3	78.4	81.5	60.3
RIT_1	37.5	77.9	91.5	73.4	18.0	94.0	49.3	45.9	82.5	81.6	63.3
NANJ2	62.0	**88.8**	91.2	66.7	40.7	93.6	42.6	**55.9**	82.6	85.2	69.3
WhuY4	42.5	82.7	91.4	74.7	**53.7**	94.3	53.1	47.9	82.8	84.9	69.2
Ours	**69.7**	84.3	**93.0**	**78.8**	37.4	**95.5**	**61.8**	46.6	**83.7**	**85.9**	**72.3**

TABLE VII

QUANTITATIVE RESULTS OF THE PROPOSED DA-NET AND OTHER DEEP LEARNING METHODS ON THE VAIHINGEN DATASET. NOTE THAT EACH COLUMN REPRESENTS THE F1 SCORE OF EACH CATEGORY, AND THE LAST TWO COLUMNS ARE OA AND AVERAGE F1 SCORE (%)

Categories	Powerline	Low vegetation	Impervious surfaces	Car	Fence/Hedge	Roof	Facade	Shrub	Tree	OA	Avg. F1
PointNet++	57.9	79.6	90.6	66.1	31.5	91.6	54.3	41.6	77.0	81.2	65.6
PointSIFT	55.7	80.7	90.9	77.8	30.5	92.5	56.9	44.4	79.6	82.2	67.7
PointCNN	61.5	82.7	91.8	75.8	35.9	92.7	57.8	49.1	78.1	83.3	69.5
PointCNN + A-XCRF	63.0	82.6	91.9	74.9	39.9	94.5	59.3	**50.8**	82.7	85.0	71.1
KPConv	63.1	82.3	91.4	72.5	25.2	94.4	60.3	44.9	81.2	83.7	68.4
RandLA-Net	68.8	82.1	91.3	76.6	43.8	91.1	**61.9**	45.2	77.4	82.1	70.9
DGCNN	67.6	80.4	90.6	54.5	26.8	89.8	48.8	41.5	77.3	81.0	64.1
GADH-Net	75.4	82.0	91.6	77.8	**44.2**	94.4	61.5	49.6	82.6	84.5	**73.2**
GACNet	62.8	81.9	90.8	69.8	25.2	91.4	56.2	39.5	76.3	81.7	66.0
GACNN	**76.0**	81.8	**93.0**	77.7	37.8	93.1	58.9	46.7	78.9	83.2	71.5
Ours	69.7	**84.3**	**93.0**	**78.8**	37.4	**95.5**	61.8	46.6	**83.7**	**85.9**	72.3

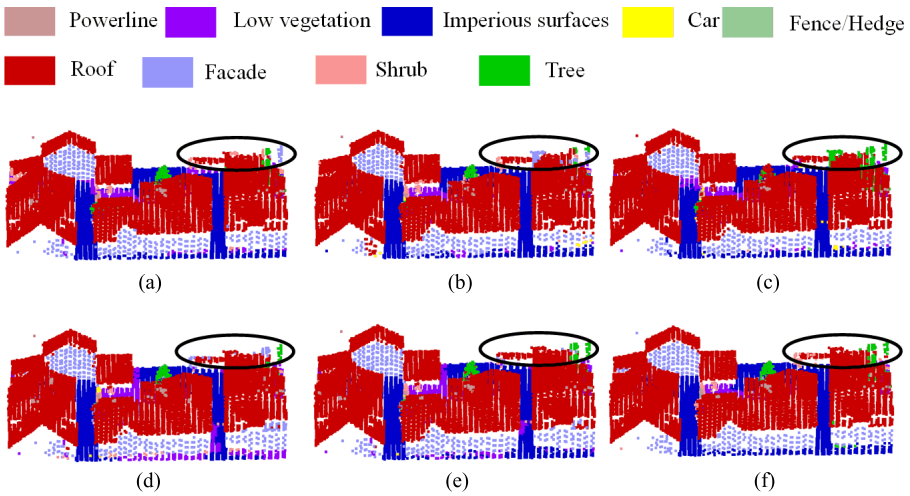


Fig. 10. Qualitative results of different models in a selected scene. Classification results of (a) LUH method, (b) NANJ2 method, (c) WhuY4 method, (d) DANCE-NET, and (e) our method. (f) Ground-truth labels. Colors indicate different categories. The area marked by the black ellipse is where our method performs better than other methods.

and the proposed DA-Net in a selected scene are depicted. When the scene contains a mix of roofs and trees [as illustrated in the black ellipse in Fig. 10(f)], the proposed DA-Net can accurately distinguish both categories [the black ellipse in

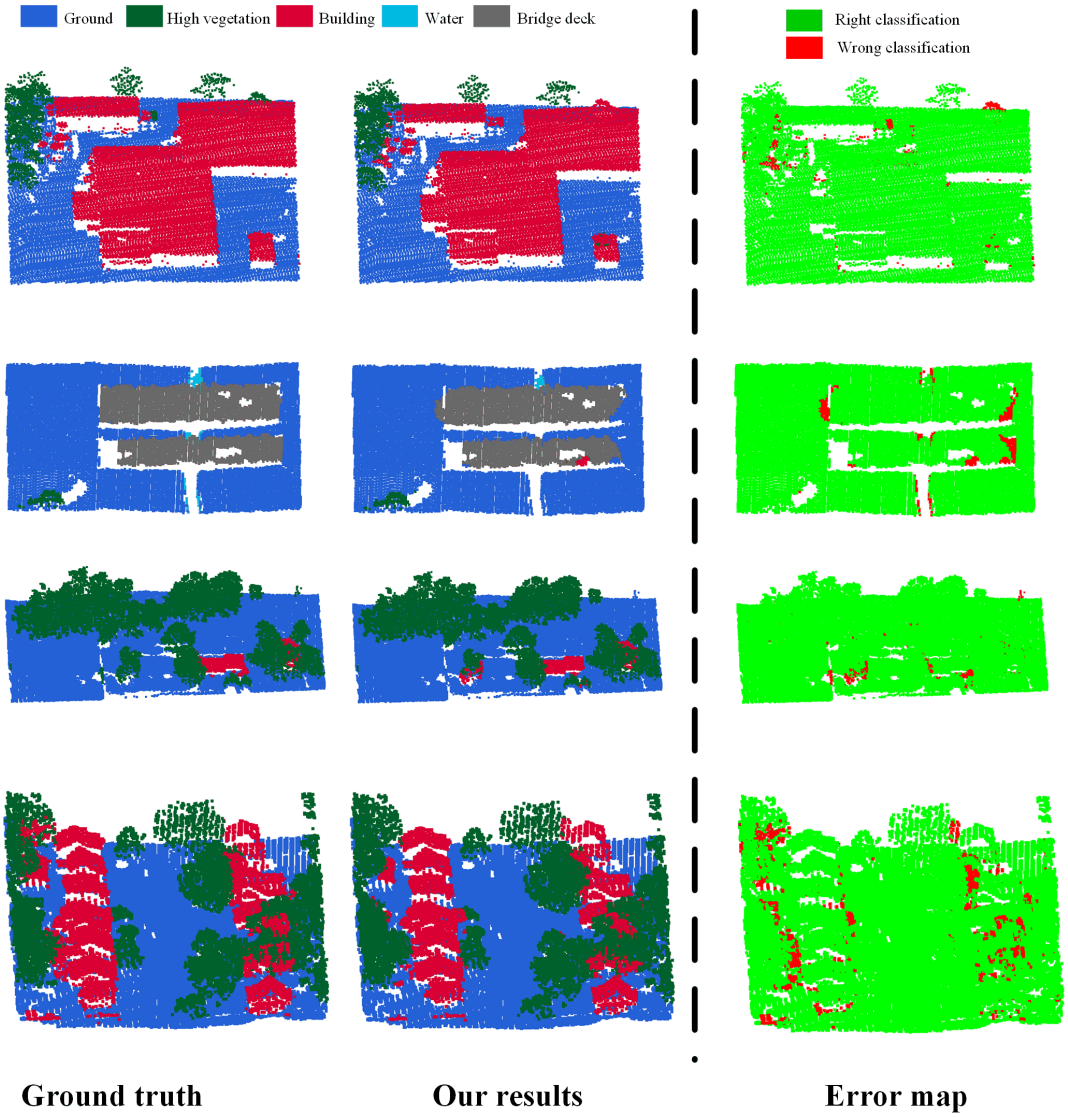


Fig. 11. Examples of segmentation results on the 2019 DFC dataset.

Fig. 10(e)], whereas the other four models have different misclassifications, which proves that it is challenging to recognize categories with similar elevation distributions [the black ellipse in Fig. 10(a)–(d)]. Besides, there is no apparent geometric structure feature to distinguish the two categories (roof and tree). An important reason that the proposed method shows better performance in the tree and roof categories is that the location-homogeneity module can well recognize elevated objects by learning rich contextual information. By contrast, the other four methods either only take into account the extraction of local structural features or perform projection transformation, leading to misclassification.

TABLE VIII
SAMPLE CLASS DISTRIBUTION OF THE 2019 DFC DATASET

Categories	Training data	Test data
Ground	48069788	6148920
High vegetation	10775292	1321488
Building	10269771	782455
Water	1356899	19266
Bridge deck	934545	25034
Sum	71406295	8297163

B. Experiments on the Data Fusion Contest Dataset

A further experiment on the 2019 DFC dataset[2] is conducted to test the generalization of the proposed method [60]. This dataset is provided by IEEE Geoscience and Remote Sensing Society (GRSS) and covers a more extensive survey area by airborne LiDAR. Therefore, it has been used for large-scale point cloud classification tasks. This DFC dataset covers Jacksonville, Florida, Omaha, and Nebraska, and is provided in the form of ASCII text files with fields *x, y, z*, intensity, and return number. Each point is to be classified into six categories:

[2]https://ieee-dataport.org/open-access/data-fusion-contest-2019-dfc2019

TABLE IX
CONFUSION MATRIX OF THE PROPOSED DA-NET ON THE VAIHINGEN DATASET WITH PER-CLASS PRECISION, RECALL RATE, AND F1 SCORE (%)

Categories	Ground	High vegetation	Building	Water	Bridge deck
Ground	6135532	932	7488	1915	2611
High vegetation	9655	1290866	18348	38	6
Building	45199	35347	695495	0	2998
Water	13677	0	0	5575	0
Bridge deck	1649	0	524	0	22306
Prcision	98.9	97.3	96.4	74.1	79.9
Recall	99.8	97.9	89.3	29.0	91.1
F1 score	99.3	97.6	92.7	41.6	85.1

TABLE X
QUANTITATIVE RESULTS OF THE PROPOSED DA-NET AND OTHER DEEP LEARNING METHODS ON THE 2019 DFC DATASET

Categories	Ground	High vegetation	Building	Water	Bridge deck	OA	Avg. F1
PointNet++	98.3	95.8	79.7	4.4	7.3	92.7	57.1
PointSIFT	98.6	97.0	85.5	**46.4**	60.4	94.0	77.6
PointCNN	97.7	97.2	84.9	44.1	65.3	93.8	78.0
KPConv	98.4	94.2	87.4	43.0	77.5	94.5	80.1
RandLA-Net	99.1	97.5	89.5	43.9	73.7	95.8	80.7
DGCNN	99.2	96.9	91.1	43.4	62.9	97.9	78.7
GADH-Net	**99.5**	97.4	92.1	34.9	**88.0**	**98.6**	82.8
GACNN	99.3	96.8	91.1	42.5	84.4	95.1	82.8
D-FCN	99.1	98.1	89.9	45.0	73.0	95.6	81.0
Ours	99.3	**97.6**	**92.7**	41.6	85.1	98.3	**83.3**

ground, high vegetation, building, water, bridge deck/elevated road, and unclassified. In addition, 100 tiles out of 110 were used as the training set, and the rest were used as the test set. The number distribution of each category of the dataset is shown in Table VIII.

For a fair comparison, the preprocessing pipeline provided by IEEE GRSS is applied to prepare the training set and the test set. Specifically, the Point Data Abstraction Library (PDAL) is first employed to divide each region of 512 × 512 m into four overlapping blocks of 512 × 512 m, 384 × 512 m, 512 × 384 m, and 384 × 384 m. Subsequently, voxels with a resolution of 1 m are used to downsample the original point clouds to reduce the amount of data and improve processing efficiency. Finally, the four blocks are chipped into several nonoverlapping smaller blocks with a max number of 65 536 points. During the training process, we randomly sample 8192 points from each small block and feed them into the proposed DA-Net. In the inference stage, the test set is processed in the same steps. To allow each point in the test set to be predicted by the network as much as possible, three slight angle rotations are performed on the raw point clouds to create three times the amount of data. Finally, the label of each point is generated by applying a nearest neighbor search method. Except that the extra input RGB features of each point are replaced by the intensity and return number, other network parameter configurations are consistent with the previous experiment.

1) Classification Results: Table IX and Fig. 11 reflect the segmentation results of our method on the DFC dataset. The results show that the proposed dual attention neural network model can better identify four categories of ground, high vegetation, buildings, and bridges. However, the proposed model performs poorly in this category because of the similarity of the water body to the ground geometry. It can be seen that the F1 scores of five out of the nine categories are more than 85%.

2) Comparisons With Other Methods: Next, the proposed method is also compared with deep learning models (PointNet++, PointSIFT [25], PointCNN [24], KPConv [47], RandLA-Net [78], DGCNN [53], GADH-Net [33], GACNN [54], and D-FCN [35]) on the IEEE 2019 DFC dataset, and the quantitative results are shown in Table X. The proposed DA-Net achieves the best performance with an average F1 score of 83.3%. In addition, it improves the best results of other models in high vegetation and building categories by 0.1% and 0.6%, respectively. The F1 scores of high vegetation and building by RandLA-Net are slightly

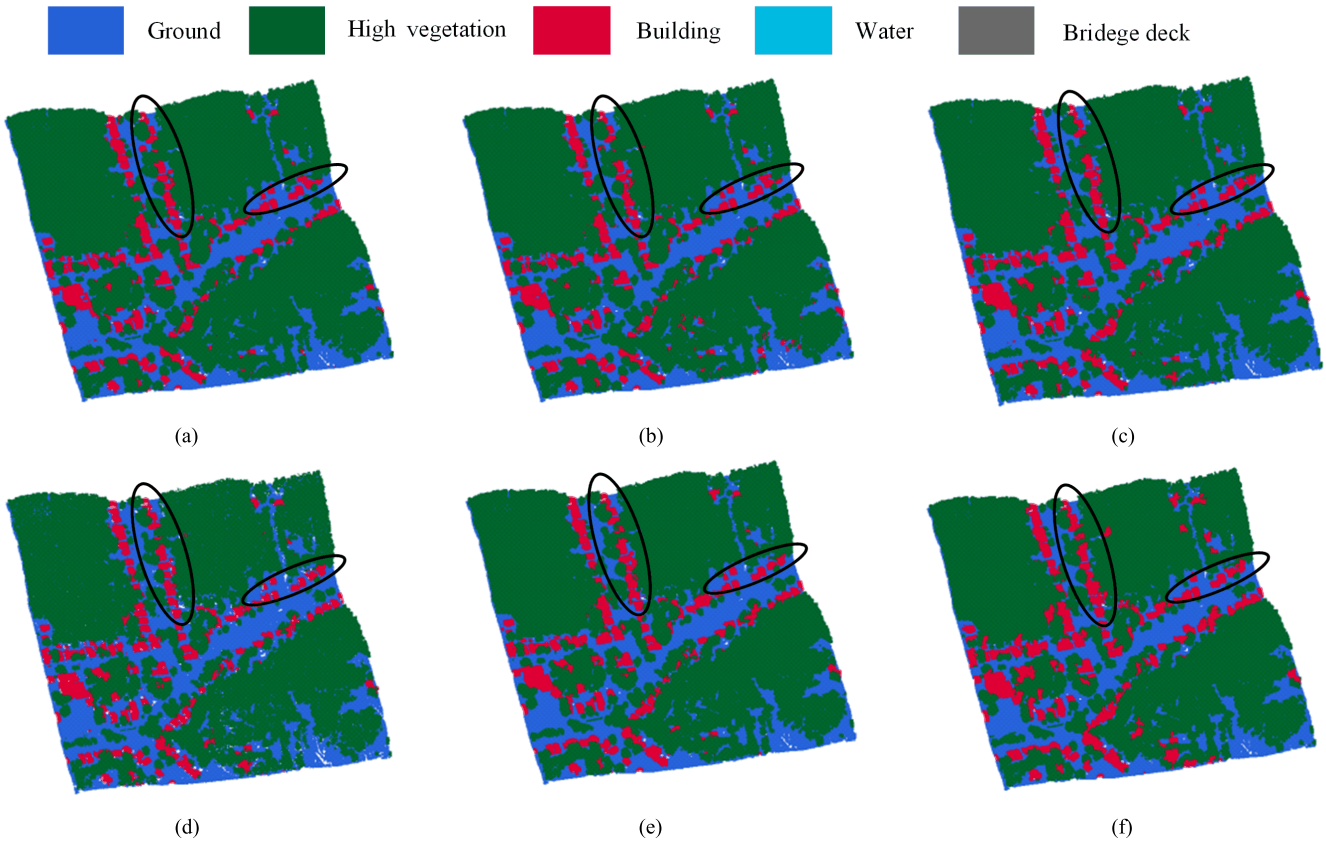


Fig. 12. Visualization results of different models in a complex scene chosen from the 2019 DFC dataset. Classification results of (a) DGCNN model, (b) DFCN model, (c) RandLA-Net model, (d) GADH-Net model, (e) proposed DA-Net, and (f) Ground truth. Colors indicate different categories. The area marked by the black ellipse is where the proposed DA-Net performs better than other models.

lower than ours since it discards key information during random sampling. It is worth mentioning that the OA (98.3%) of our method is slightly lower than GADH-Net [33] due to the introduction of the weighted cross-entropy loss function. For visual comparison, the results of other models, including DGCNN [53], D-FCN [35], RandLA-Net [78], and GADH-Net [33] in a selected 512 m $\times$ 512 m region, are illustrated in Fig. 12. It can be seen that there are misclassifications of building points as high vegetation in the other models (the black ellipse in Fig. 12). In contrast, the proposed DA-Net can identify points in the above two categories correctly, especially on the edge of building roofs. A possible reason is that these models cannot effectively capture long-range contextual features. Specifically, DGCNN [53] uses Edge-Conv to obtain local geometric features but lacks encoding of context information, while D-FCN [35] designs a directional constrained full-winding machine network to encode directional information in the local receptive field, which, however, ignores the correlation of features outside the receptive field. Compared with RandLA-Net [78], our model is also better at preserving the details of buildings. The GADH-Net [33] achieves the best performance in the ground category. In contrast, the proposed DA-Net achieves better performance than the other models due to the effectiveness of the LH that can selectively aggregate contextual information, thereby keeping semantic consistency in the same object.

V. Conclusion

In this article, a DA-Net is proposed for semantic segmentation of airborne LiDAR point clouds. To better obtain local topological structure features, a DA module is designed, composed of an AER and an EAP. Between them, the AER can construct a spatial representation that is robust to orientation and position. The EAP focuses on the elevation information to enhance local structural features. Finally, an LH module is designed to explore long-range dependencies. By recursively employing the DA and LH, an end-to-end architecture that can be directly applied to the raw LiDAR point cloud is constructed.

Furthermore, we conducted a series of experiments to verify the effectiveness and robustness of the proposed method, which has achieved state-of-the-art results on the ISPRS Vaihingen dataset in terms of both OA and average F1 scores. Results show that, even without HaG features as input, the proposed method outperforms many recent deep learning methods. We further carried out experiments on the 2019 DFC dataset to demonstrate the generalization ability of the proposed model in large-scale airborne LiDAR point cloud classification.

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